GAMEPLAY TAGS

Welcome!

NEWS: [Join the brand new Telepresent Games Discord Server!]

Thank you for choosing the Gameplay Tags! I hope it'll be a nice addition to your toolbelt!

In this documentation, you'll find everything you need to get started, from installation to feature explanations and usage tips

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Let's dive in!

Why Gameplay Tags?

Gameplay tags provide a flexible and powerful way to organize and manage game data in Unity. By using a hierarchical system of meaningful labels, developers can easily group objects, abilities, and states, streamlining condition checks and reducing reliance on complex hard-coded logic. For example, instead of using multiple if-statements to determine whether a character is an enemy, a simple tag check like `if (IsTagActive("EnemyTag"))` can be used, which improves both code clarity and maintainability.

Moreover, gameplay tags seamlessly integrate across different game systems—from AI behavior and combat mechanics to UI filtering and level design triggers. As ScriptableObjects, they allow for easy expansion and modification without the need to rewrite existing code, ensuring that new features can be added with minimal disruption. This

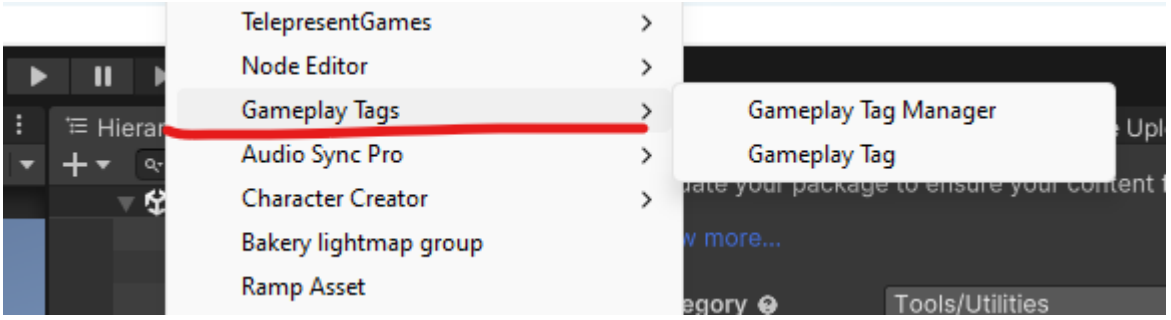
unified approach not only enhances the scalability of your project but also fosters a cleaner, more organized development environment.

Installation

Not much to say here! Simply import the asset from the Package Manager, and you'll gain immediate access to the Gameplay Tag and Gameplay Tag Manager systems!

How To Use

Once installed, you'll be able to add Gameplay Tags and Gameplay Tag Managers using the right click menu in your project view.

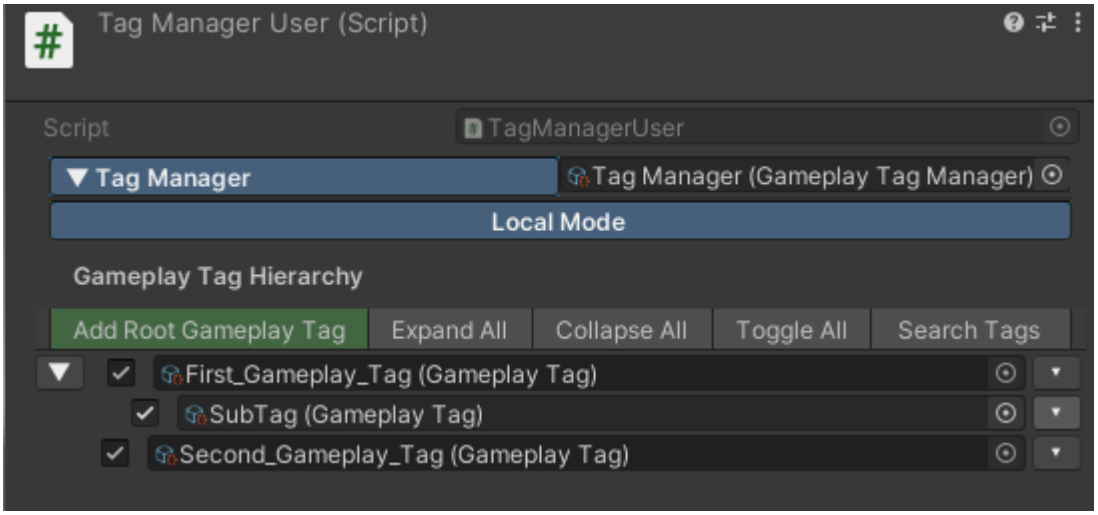


To use the system in your game, you may add Gameplay Tags and Gameplay Tag Managers to your game objects. This will allow you to use GamePlay Tags as a way to streamline conditional logic in your project, and decoupling scripts, improving scalability.

Gameplay Tag

The Gameplay Tag is a scriptableObject, which has a TagActive bool. A Gameplay Tag can be added to an object as a sole Tag, or as a part of a Gameplay Tag Manager Hierarchy.

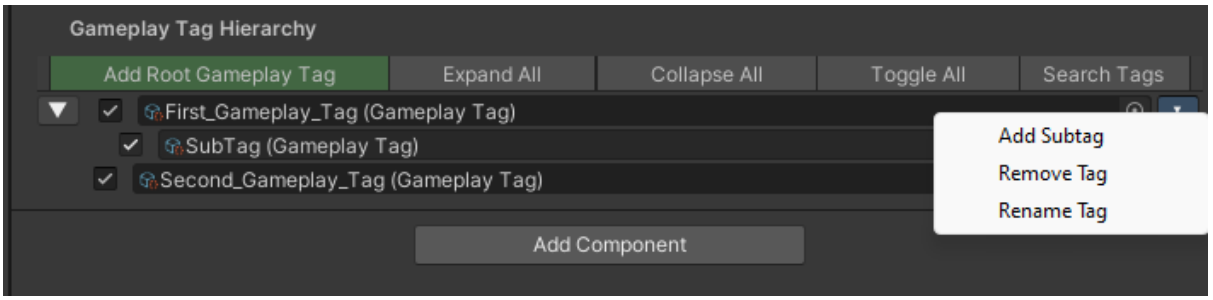
Gameplay Tag Manager



The Gameplay Tag Manager is a ScriptableObject, which allows you to create a hierarchical index of Gameplay Tags. When using a Tag Manager, you can add any amount of sub-tags to a Gameplay Tag.
When attached to a GameObject, the Tag Manager also has two modes to switch between:

- **Global Mode:** When in Global Mode, any changes made to a Gameplay Tag in the Manager will also carry over to the Gameplay Tag ScriptableObject itself. Conversely, any change made on a Gameplay Tag will also instantly appear in the Manager.
- **Local Mode:** Any changes made to a Gameplay Tag's active state stays within the Tag Manager, and will not affect the overall Gameplay Tag ScriptableObject. This is particularly useful for cases where you may for example have multiple Objects referencing the same Manager.

When queried from a Tag Hierarchy, a subtag's parent tags all need to be active before that child-tag can be recognised as active.



In the Tag Manager, you can also find many controls for managing individual Tags, as well as some useful utilities for the Tag Manager overall.

Gameplay Tag Scripting Overview

GameplayTag.cs Overview

GameplayTag is the main class for individual Gameplay Tags and primarily consists of a simple boolean state.

- **TagActive (bool):**
  Indicates whether the Gameplay Tag is active.

- **SetTagState(bool state):**
  Sets the TagActive bool of the Gameplay Tag.

- **ToggleTagState():**
  Toggles the TagActive bool of the Gameplay Tag.

- **IsTagActive(GameplayTag tag):**
  Checks if a standalone GameplayTag asset is active. Use this when not working within a manager context.

- **OnTagChanged (Event):**
  An event that can be subscribed to. It is called whenever the tag state changes.

GameplayTagManager.cs Overview

The GameplayTagManager holds a hierarchy of GameplayTags. Most methods related to the manager can be found in GameplayTagUtility.cs.

- **OnManagerTagChanged (Event):**
  An event that can be subscribed to. It is called whenever a specific tag in the Manager changes.

- **OnAnyManagerTagChanged (Event):**
  An event that can be subscribed to. It is called whenever any tag in the Manager changes.

GameplayTagUtility.cs Overview

GameplayTagUtility is a static class that provides a range of utility methods for checking and toggling the state of GameplayTags, both in isolation and within a hierarchy managed by a

GameplayTagManager. These methods streamline common tag operations in your game logic.

- **IsTagActive(GameplayTag tag):**
  Checks if a standalone GameplayTag asset is active. Use this when not working within a manager context.

- **IsTagAndParentsActive(GameplayTagManager manager, GameplayTag tag):**
  Determines if the specified GameplayTag is active within a GameplayTagManager hierarchy by ensuring that the tag and all its parent tags are active.

- **IsTagAndParentsActive(GameplayTagManager manager, string tagName):**
  Similar to the previous method but first looks up the tag by its name in the GameplayTagManager before verifying its active state along with its parents.

- **IsTagLocallyActive(GameplayTag tag):**
  Returns the opposite of IsTagActive for a standalone GameplayTag, indicating if the tag is inactive.

- **GetSubtagsFromManager(GameplayTagManager manager, string tagName):**
  Retrieves a list of all child GameplayTags (subtags) associated with the specified parent tag name in the manager.

- **ToggleTagState(GameplayTag tag, bool active):**
  Toggles the active state of a standalone GameplayTag asset to the desired state.

- **ToggleLocalTagState(GameplayTagManager manager, GameplayTag tag, bool active):**
  Toggles the active state of a GameplayTag through the GameplayTagManager, applying the manager’s logic (including local mode handling) to update the tag's state.

Thank You for Using Gameplay Tags!

I hope this tool makes it more fun and smooth for you to work with conditional logic in your project!.

If you have any questions, feedback, or run into any issues, please don't hesitate to reach out at **TelePresentGames@gmail.com**, or join the Discord 🙂 Your input is invaluable.

Happy developing!

Kind regards,
Martin

Changelog:

Original Release:

- Original release.